

TOPOLOGICAL ANALYSIS AND SIMULATION OF A DYNAMIC PROCESS IN A REAL NETWORK

STUDY OF SCIENTIFIC COLLABORATIONS AND HOW AN OPINION MAY SPREAD

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Abstract

Real network analysis is widely utilized in various fields, including biology, technology, and social sciences. These networks exhibit an intriguing property known as the “small world” phenomenon. This property enables these systems to exhibit both high clustering, akin to regular networks, and short characteristic path lengths, akin to random graphs. To gain insight into the structure of such networks and to assess their small-world nature, a real network was selected from a repository. This network contains data on scientific collaborations in the field of networks. By computing its topological features and examining the presence of the small-world property, this selected network served as a basis for comparison with two models: the small-world model and the configurational model. Through this comparative analysis, the characteristics of the reference network were evaluated and contrasted with the characteristics of the two models. Furthermore, we employed a dynamic process, specifically the SIS model, to simulate the propagation of opinions or information through infected and susceptible nodes within the network. This dynamic modeling approach enabled us to explore how the small-world structure influences the spread of opinions and the dynamics of information diffusion.

I INTRODUCTION

Complex networks have attracted a great deal of attention in various fields due to their efficient representation of a wide range of systems across different disciplines. Biology, for example, uses protein interaction networks. Similarly, computer science relies on these networks to analyse the World Wide Web (WWW) and the Internet. In addition, the social sciences use complex networks to study collaboration, communication, economic and political networks. Moreover, not only the networks originate from different domains, but the methodologies of network science as well. For instance, it uses theories and methods from disciplines such as graph theory, statistical physics, computer science, statistics and sociology.

More recently, networks have gained prominence in physics and other fields, providing a mathematical basis for representing complex systems. Topology, in particular graph theory, has made it possible to analyse these networks, and some common features of many networks have come to light, such as the sparse property, the small

world feature, clustering and community structure. Some of these properties, their calculation and the code used to obtain them are presented, with the results shown in section II. The real network chosen for the study is available at the The Colorado Index of Complex Networks (ICON) [1] repository under the name “Scientific collaborations in network science (2006)” and it has been used at reference [2]. Also, a new and extended version is used at reference [3]. Due to the nature of the network, some assumptions can be made beforehand; for example, it is highly unlikely that there was a hierarchy where one node is connected to almost all other nodes, and on the other hand, a high degree of local clustering is expected. The former is because the number of collaborations and publications is limited in a career duration, with the possible exception of Paul Erdős, and the latter because collaborations are more likely to take place in a particular group of research.

On the other hand, some network models than set of rules to interconnect a number of elements in order to reproduce specific features has been developed. Those models play an essential role in understanding the structural properties and dynamics of complex networks, providing

valuable tools for analysing real-world systems in various domains. For instance, the “small world” model developed by Watts and Strogatz [4] in 1998 is a model that focuses on networks that have both high clustering and short characteristic path lengths, resembling a combination of regular lattices and random graphs. This model provides insights into how information or influence can spread efficiently through the network due to its compact structure. With a similar objective but in a totally different approach, the configurational model is a theoretical framework used to study network properties by comparing them to an ensemble of randomised networks with preserved structural properties. By creating randomised versions of a given network while preserving its degree distribution or other relevant features, the configurational model helps to assess whether observed network properties are the result of specific structural arrangements or simply a consequence of the size and degree distribution of the network. There are other network models, such as the random graph of Erdős and Rényi or the rewiring, but they will not be explained in detail as they are not used in this work.

Networks offer much more than just the analysis of their structural features. They serve as a rich field for studying dynamic processes, which encompass phenomena unfolding over time within the network structure. For instance, the spread of epidemics, where infected individuals can only infect their neighbors with a certain probability, is modeled by the SIS (Susceptible-Infected-Susceptible) model. This model is not only used to understand the transmission of infectious diseases but also rumors and opinions within a population. Therefore, it will be used to simulate and analyse the spread of a certain opinion or idea between the authors that collaborate. The SIS model can be formulated as a Langevin equation, so stochastic simulations using the Gillespie algorithm are utilized. Through this approach, a comprehensive understanding of the dynamics of spreading phenomena in networks can be achieved.

This work is organized as follows: in the first part the selected real network is analysed and its topological properties are computed. Next, the “small world” model and the configuration model are used as a comparison to understand those properties of the real network. For the “small world” model the comparison will be done by making the networks as the same number of nodes and the configuration model uses the degree distribution of the network.

II FEATURES OF THE CO-AUTHORSHIP NETWORK

Network analysis is a valuable tool for understanding connectivity and other properties of networks. This discipline, rooted in graph theory, falls under the field of topology. In this project, we focus on a simple network that is

unweighted, undirected, and not embedded in any specific space. Therefore, the only metric available for this network is the number of links.

The network we examine is the co-authorship network of network scientists. Each node represents an author who has published at least one paper in network science, and two nodes are connected if the corresponding authors have collaborated on at least one network science paper. Since the network is unweighted, there is no measure of the strength of the connection between two scientists based on the number of joint papers. The network has 1589 nodes, although connected to the network are only 1461, and 2742 edges with an average degree of 3.45, similarly, the median degree is 3. The largest connected component consists of 379 nodes and it is depicted in Fig. II.1

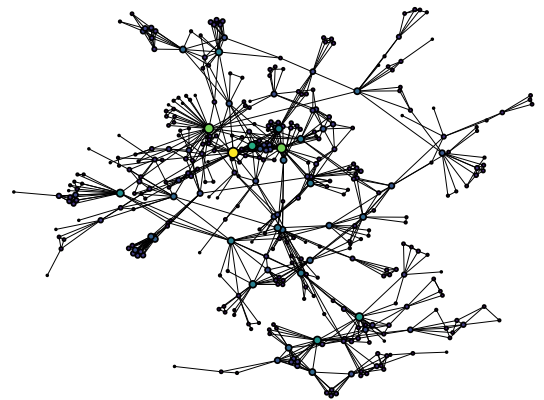


FIGURE II.1: The largest connected component of the co-authorship network of network scientists

The degree distribution of the network is depicted in Fig. II.2. Within the graph, there are 128 isolated nodes, as seen in Fig. II.2a, which corresponds to scholars with a single-authored network science paper who haven’t co-authored any other network science papers. The majority of co-authors fall within the range of 1 to 3, with the highest degree being 34 attributed to A. Barabasi, well-known scientist of network science known among other achievements for doing the network model with preferential attachment. Also, in Fig. II.2b the degree distribution $P(k)$ is shown in comparison with the expression of $P(k) \propto k^{-\gamma}$ if k is taken as a continuous variable $P(k) = \frac{\gamma-1}{k_{min}} \left(\frac{\gamma-1}{k_{min}} \right)^{-\gamma}$ and using an exponent $\gamma = 2.5$. This choice of γ is motivated because real network usually has that exponent since they are heterogeneous and real networks follow a fat-tailed power-law distribution and that is why they are called scale free.

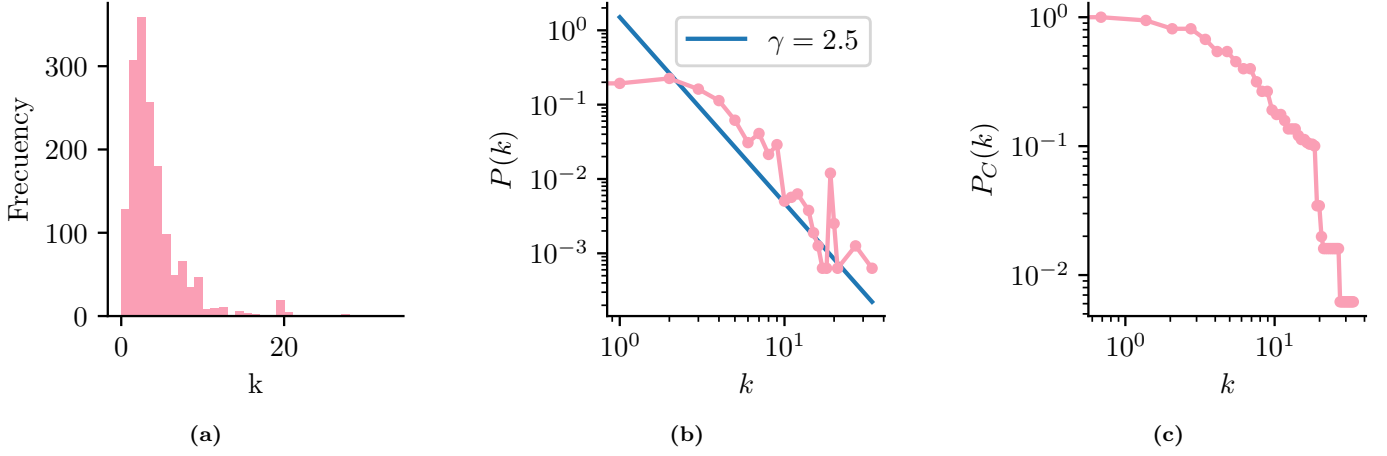


FIGURE II.2: a) Histogram of the frequency of each degree of the network. b) Degree distribution of the network. c) Complementary cumulative degree distribution.

The network has a high assortativity coefficient of 0.46 that suggests that nodes tend to be connected to other nodes with similar degrees. The average nearest neighbors degree (ANND), that is a measure of the tendency of nodes to connect to peers in terms of degree, is shown in Fig. II.3.

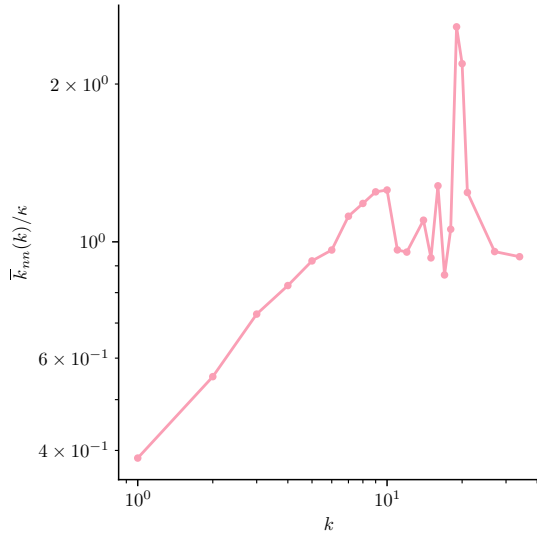


FIGURE II.3: The average nearest neighbors degree of the network. It is assortative mixing.

From the growing of this function it is concluded that the network is assortative mixing, so there is a correlation of nodes of similar degree. That is usually the case for social networks, as it is the co-authors network, because nodes tend to be connected with other nodes with similar degree values. To visualize this feature, the k-cores of the network have been calculated and presented in Fig. II.5, illustrating that nodes within the same k-core are often interconnected, forming local clusters.

The co-authorship network is highly clustered with a global clustering coefficient of 0.693 and a average local clustering coefficient of 0.520. The degree dependant clustering coefficient is represented in Fig. II.4. As shown in the figure, the network reveals significant clustering, particularly among nodes with low degrees. Additionally, there is a peak at a high degree of $k = 20$, indicating that neighbors of nodes with degree $k = 20$ are also highly interconnected. This behavior may be attributed to the specific construction of the network, as well as its assortative nature, suggesting strong interconnections among nodes with degree $k = 20$. This observation is further supported by the peak in the function ANND at the same degree. Lastly, the average shortest path length of 6.04 in the largest connected component reinforces the small-world nature of co-authorship networks.

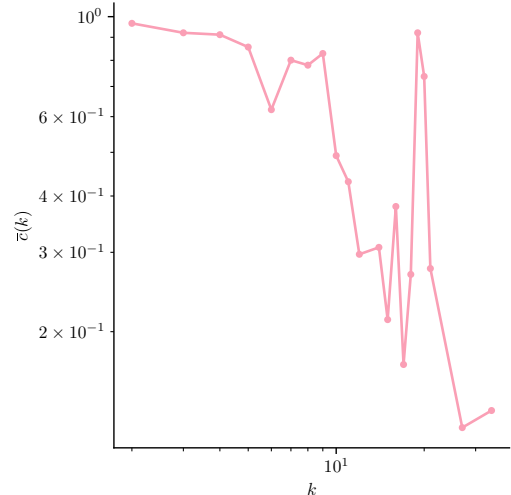


FIGURE II.4: The degree dependant clustering coefficient of the network. It heavily clustered for nodes of low degree.

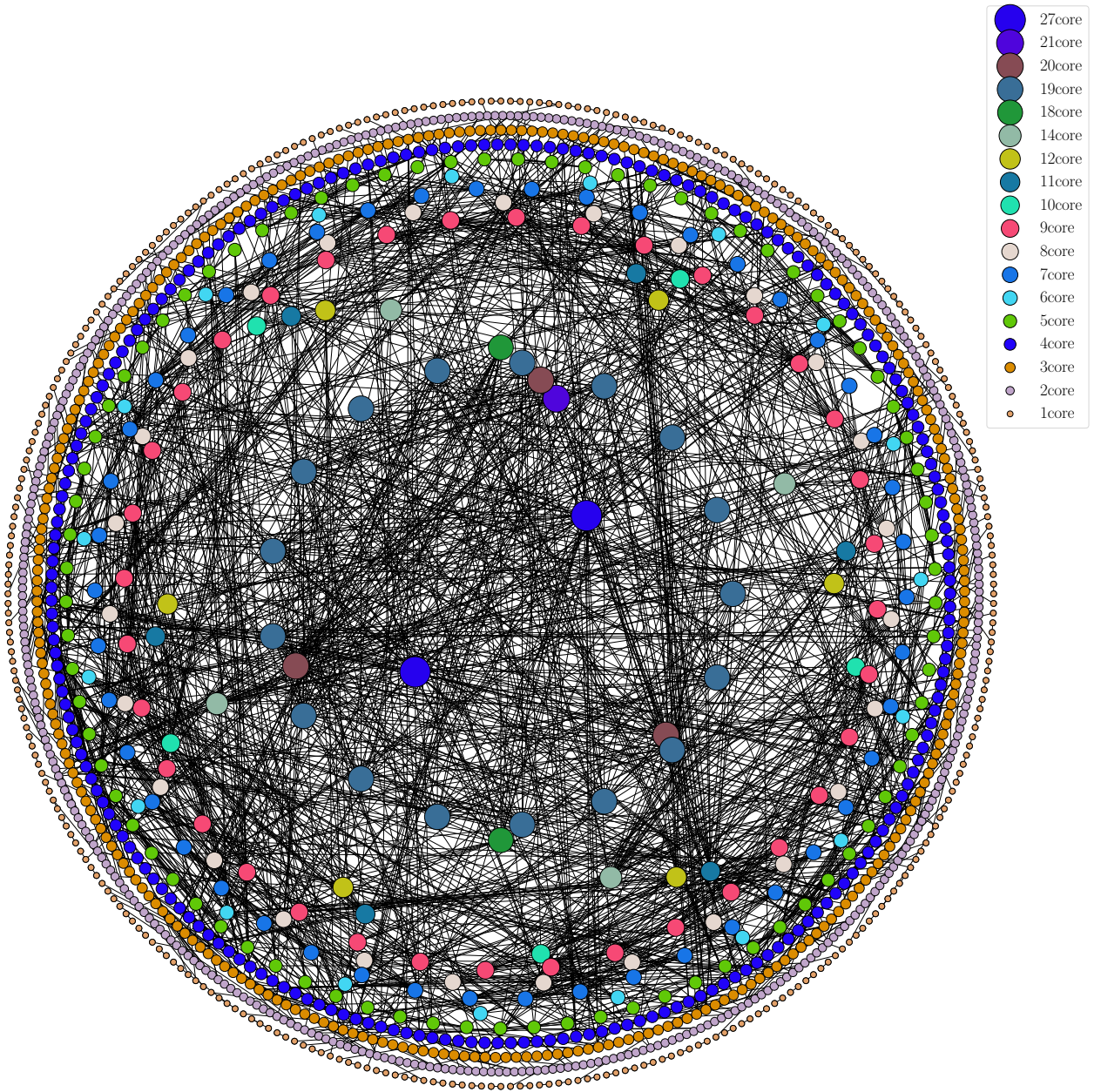


FIGURE II.5: K-core representation of the network.

III COMPARISON OF A REAL NETWORK WITH SW AND CM MODELS

The comparison between models will be done with small world and the configuration model. Those two models are designed to explain different features each. The small world model how networks with small-world properties can arise in various domains. It also shed light on phenomena like the “six degrees of separation”, a name popularized by Milgram’s experiment [5], where any two individuals in a large population can be connected through a relatively small number of intermediaries. On the other hand, the

configuration models tries to check if a random graph that preserve some features of the reference network has comparable topological properties.

To compare the co-authorship network to the small-world model, we use the same number of nodes (1589) and set the average degree to be comparable. Therefore, the network done by SW will have 1589 nodes and $\langle k \rangle$ will be taken as $\langle k \rangle = 4$ as it is comparable to our network. Initially, the model starts with a regular lattice network, where each node is connected to its k nearest neighbors by undirected edges. For choosing the edges, an easy way to create those links is using a circular shift of the nodes arrays and make the links between the shifted and non-shifted nodes. For connecting it to the k nearest neighbors

the circular shift scheme has to be done $k/2$ times. That is why $\langle k \rangle = 4$ was chosen.

To introduce the small-world effect, the model randomly rewires some edges of a regular lattice. This rewiring process involves selecting an edge and reconnecting it, with some probability p , to a randomly chosen node in the network. By adding these random connections, the model creates shortcuts or long-range connections that enable shorter path lengths between distant nodes.

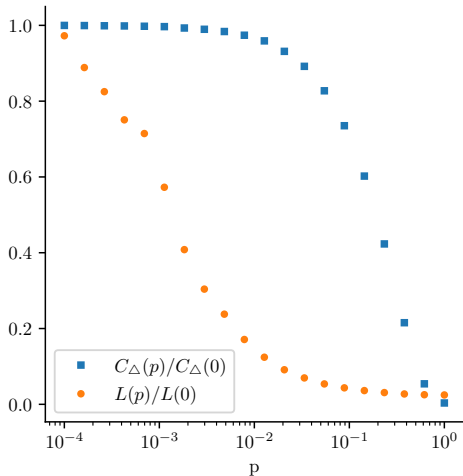


FIGURE III.1: Small world model. The square markers are the global clustering coefficient while the circular marker is the average shortest path length. Axis x is the probability of rewiring and edge, p . To smooth the curve the computation has been done with twenty repetitions for each value of p .

The small world model demonstrates that a relatively small number of random connections is enough to significantly reduce the average path length between nodes while maintaining a high level of clustering (that will be the case for the central values of p in Fig. III.1).

The configuration model is usually used as a null model for real complex networks. It produces maximally random networks with a preassigned degree sequence that is uncorrelated except for structural unavoidable correlations necessary to close the networks. Null models are important in the study of complex networks because they provide a baseline for comparison, assess the statistical significance of network features and identify deviations from random expectations. It is a stochastic algorithm that constructs a network by randomly rewiring the edges while preserving the prescribed degree distribution. They help differentiate between random variations and meaningful patterns, enabling to understand the underlying dynamics and processes that shape complex systems.

The methodology for comparing the real network with

the configuration model is using the distribution of degrees calculated in the real network and use it for obtaining a random network with the same degree distribution, but nearly uncorrelated and that not reproduce clustering. From the new network, topological features are calculated and the process is repeated several times to make possible some statistics.

The configurational model for networks begins by defining a set of nodes with their desired degrees. The total number of edges is calculated based on the degree sequence. A pool of stubs is created, where each node contributes stubs proportionally to its degree. The algorithm iterates until all stubs are exhausted: it randomly selects two stubs from the pool, connects them if there is not multiconnections or self-loops, and removes the selected stubs. Finally, the resulting network is obtained when all stubs have been used.

The topological properties has been computed for a hundred of networks make by Configuration model, using for all of them the degree distribution of the real network. For all those repetitions the average of the topological properties has been done. In Fig. III.2 is depicted for comparison the ANND function of the real network and that of the hundred of repetitions with CM. The analysis reveals that the correlation between degrees in the real network disappears when computed for the average of the configuration model (CM). This implies that the correlation and assortativity observed in the co-authorship network are not merely a size effect but rather an inherent feature of the network itself.

On the other hand, Fig. III.3 represents the degree dependant clustering coefficient for the real network and the configurational model, the later also averaged for a hundred of repetitions. As it was expected, in the configurational model there is not presence of clustering, except for that clustering due to side effect. Furthermore, global clustering coefficient has been calculated doing the mean over the hundred of repetitions and $G_{\Delta} = 0.0064$ with almost none variation over different realizations. That value is much more lower than the global clustering of the real network.

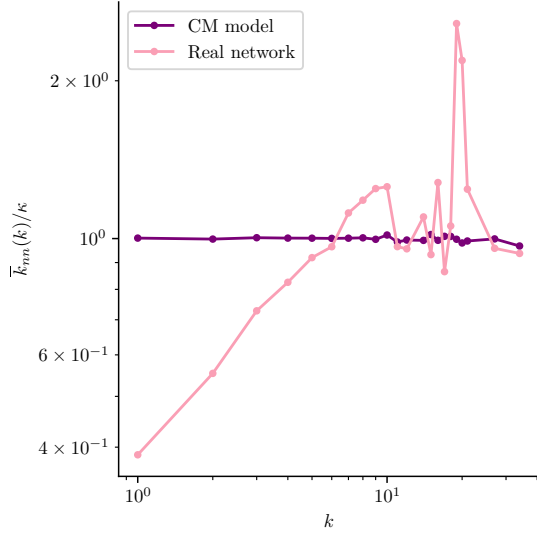


FIGURE III.2: The largest connected component of the co-authorship network of network scientists

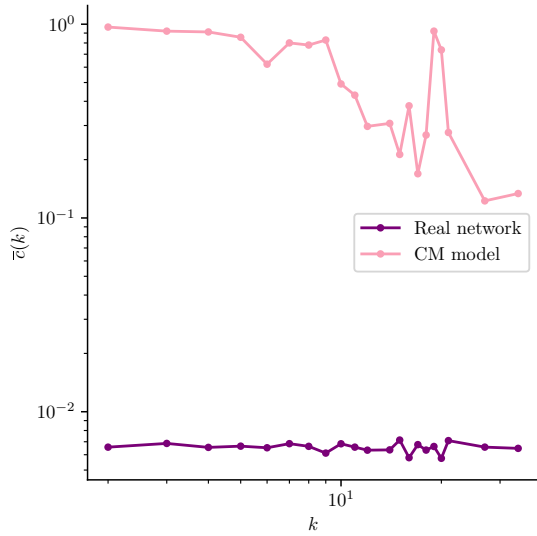


FIGURE III.3: The largest connected component of the co-authorship network of network scientists

IV DYNAMIC PROCESS: SIS MODEL

In the real network previously presented, the SIS model will be used to simulate the spread of an opinion between the authors. It will be a temporal opinion as the individual can recuperate. As it was seen in the representation of the network Fig. II.5 the network is heavily fragmented; in fact, the largest component is of 379 nodes while the network has 1589 nodes. Therefore, a giant cluster and the total percolation of the network is impossible to reach. The expected consequence of this effect is that the prevalence $\rho(\lambda)$ will have a low limit value.

The SIS model is usually used to model epidemics or transmission of information between individuals. The idea is simple, there is two possible states for each node: infected $n_i = 1$, then it can be recuperated, or susceptible $n_i = 0$, it can be infected. There is a rate for infection λ and another one for recuperation δ , since the importance parameter on the dynamics is λ/δ , δ is taken as one for simplicity. With those rates and after some derivation a differential equation is obtained:

$$\frac{d}{dt}\rho_i = -\rho_i + \lambda \sum_{j=1}^N a_{ij}\rho_j(1 - \rho_i) \quad (\text{IV.1})$$

where ρ_i is the permanence of each node and is defined $\rho_i = \langle n_i \rangle$, N is the number of nodes of the network and a_{ij} are the terms of the adjacency matrix. This equation can be solved as a differential equation, using for instance a Runge-Kutta 4, but also can be solved using the Gillespie algorithm and by means of stochastic methods. This implies running a simulation on each time step compute the probability of recuperation as $p = n_I/(n_I + \lambda E_{act})$ where n_I are the number of infected nodes and E_{act} is the number of active links, the probability of infection will be $1 - p$. Using the computed probability it is decided if a recuperation or an infection will take place. The only condition for a recuperation is that the node is infected and for a infection a infected neighbor is required. Therefore, if the mean prevalence $\rho = \frac{1}{N} \sum_i \rho_i$ reach zero the process ends and no more propagation will be possible. Each time step between two process is calculated by a exponential distribution with rate $n_I + \lambda E_{act}$.

To check the two possible methods of resolution, both will be carry out and it is expected that they give the same result. The methodology to carry out the simulation is to prepare an initial configuration with a number of nodes $n_I^0 = 50$ infected randomly. This initial condition is used to all the following calculations. With that configuration 40 different values of λ , linearly distributed from $\lambda = 0.05$ to $\lambda = 4$. In addition, the Runge-Kutta integration is done once for the initial condition and the stochastic simulation is done with ten repetitions and taken the mean value. The results of this procedure can be found at Fig. IV.1. Contrary to what was expected, the results doing the integration and those using stochastic methods do not coincide although they follow a similar behavior. That is surely due to an error in the program, but none error have been found after several revisions of the code. In the light of the results, the critical lambda parameter is approximately $\lambda \approx 0.15$. More over, the maximum value of ρ is briefly over $\rho = 0.3$ what effectively is a low value and implies that the network is not totally infected. As it was said before, this was the most likely outcome since the network is fragmented and, as it can be seen at Fig. II.5 there are a lot of pairs of nodes only connected between them.

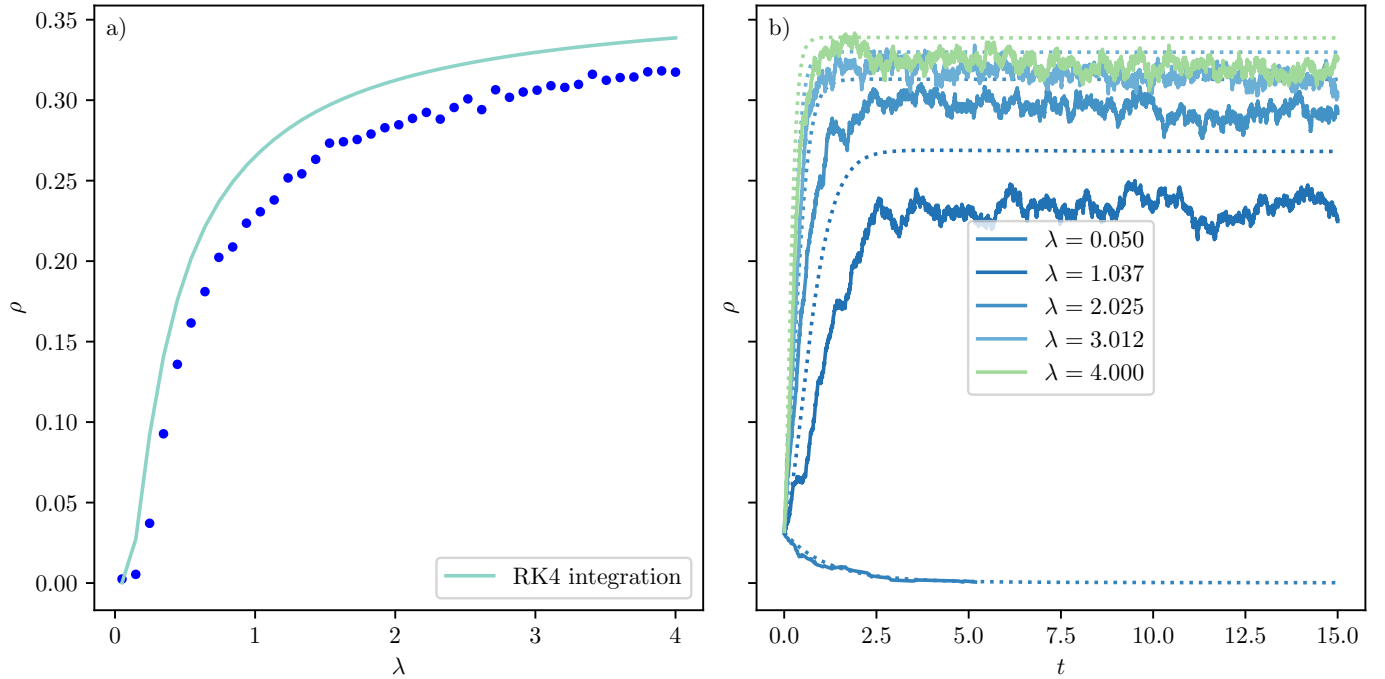


FIGURE IV.1: Results of the dynamic process with an initial condition of 50 infected nodes for different λ . a) Mean values of the steady state after sunning the different simulation, the blue points are the results obtained by the stochastic simulation. b) Time evolution of ρ where each color impy a value of λ and the dotted line is the RK integration while the solid line is the stochastic simulation.

V CONCLUSIONS

The co-authors network has been analysed and some of its topological feature such as the degree distribution, the clustering and the correlations between degrees. From that analysis it has been concluded that the co-authors network is heterogeneous and scale-free, as its degree distribution can be compared to a power-low of exponent $\gamma = 2.5$. Also, they are several isolated nodes and is highly fragmented, what makes obvious because the largest connected component has 389 nodes. For hat largest component the average shortest path length has been calculated as 6.04 edges; therefore, it fulfils the small-world condition of real networks. For the correlation between degrees, it has been found that the network is assortative mixing and also heavily clustered, not only globally but also compared with each degree class. For the clustering dependent of the degree, is have been found that the lower values of the degree has more clustering that lower values of k , except for a peak found around $k = 20$.

About the comparison of the real network with the small wold model, it has been found that a network with the same number of nodes and $\langle k \rangle$ can show a high value of clustering with a low average path length; therefore, the real network would have similar properties as a small world model network with a middle value of probability p . It is not like a regular network nor a random graph.

With the comparison of the configurational model it have been found that the correlation between nodes and the clustering is not a size effect but of the social network nature.

In the second part of the project the dynamic process have been simulated and, although the integration and the stochastic simulation does not coincide for some obscure reason, the results depicts that the critical parameter λ_c is around $\lambda_c \approx 0.15$, that is a low value that can be related with the small world property of reals networks, and the maximum value of λ barely surpass $\lambda = 0.3$, that implies that only a third part of the network ends infected. That is a consequence of the fragmentation of the network, as it is not fuled connected.

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